MIL-B-13136B
26 January 1968
SUPERSEDING
MIL-B-13136A
10 March 1960

MILITARY SPECIFICATION

BATTERIES, DRY

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single-cell dry batteries with a chloride electrolyte and having a nominal voltage of 0.9 volt. (See 6.1)

1.2 Classification.

1.2.1 Type designation. The type designation of dry batteries is derived from the AN nomenclature system specified in Publication MIL-STD-196. Dry batteries shall be of the following types (see 6.2):

BA-245/U - Suitable for use in temperate and tropical zones.
BA-2245/U - Suitable for use in arctic zones.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-B-613	Brass, Leaded and Non-leaded; Plate Rolled Bar, Sheet and Strip
QQ-B-626	Brass, Leaded and Non-leaded; Rod, Shaped, Forgings and Flat Products with Finished Edges (Bar and Strip)

FSC 6135

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QQ-C-502	Copper Rods and Shapes; and Flat Products with Finished Edges (Flat Wire, Strips and Bars)
QQ-C-576	Copper Flat Products with Slit, Slit and Edge-Rolled, Sheared, Sawed or Machine Edges (Plate, Bar, Sheet and Strip)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-S-571	Solder, Tin Alloy, Lead-Tin Alloy and Lead Alloy
QQ-T-191	Terne Sheet (Long and Short Terne)
PPP-B-566	Box, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-676	Box, Set-up
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced
L-P-378	Plastic Film (Polyethylene Thin Gage)

MILITARY

MIL-W-76	Wire and Cable, Hookup, Electrical, Insulated
MIL-P-116	Preservation, Methods of
MIL-B-857	Bolt, Nut, and Stud
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base)

STANDARDS

FEDERAL

FED STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads (40 in. x 48 in. 4-Way

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(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Threaded Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, One Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Preproduction. Batteries furnished under this specification shall be a product which has been subjected to and has passed the preproduction inspection specified in 4.5.

3.2 Materials and components. When a definite material or component is specified, it shall be in accordance with the applicable specification or requirement listed in Table I. When deemed necessary by the Government, certification from the source of the material or component will be required. In the absence of certification from the source, a certificate of analysis or certified inspection data will be required.

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Table I - Materials and components

Component material	Rqt Para	Applicable Specification
Solder [1]	---	QQ-S-571
Soldering flux [2]	---	MIL-F-14256
Metals	3.2.1	
Dissimilar metal	3.2.2	
Terneplate	---	QQ-T-191
Brass	---	QQ-B-613 or QQ-B-626
Copper	---	QQ-C-502 or QQ-C-576
Cadmium plating	---	QQ-P-416
Machine screws, studs, and nuts	---	MIL-B-857
Wire, hookup	---	MIL-W-76

[1] For electrical connections type SN 40 or higher tin content shall be used.

[2] If other fluxes are used, they shall not affect the performance of the battery or reduce the shelf life.

3.2.1 Metals. All metals which do not enter into the basic electro-chemical reaction of the cell, if not corrosion resistant, shall be plated or treated to resist corrosion, when subjected to any test or storage conditions specified herein.

3.2.2 Dissimilar metals. Metals having dissimilar electrolyte potentials which are in intimate contact with each other and in presence of moisture, under the ambient conditions specified herein, shall be protected against electrolysis and corrosion.

3.3 Design and construction. Batteries shall be of the design, construction, and physical dimensions shown on Figure 1 (see 3.3.1 to 3.3.8, incl.).

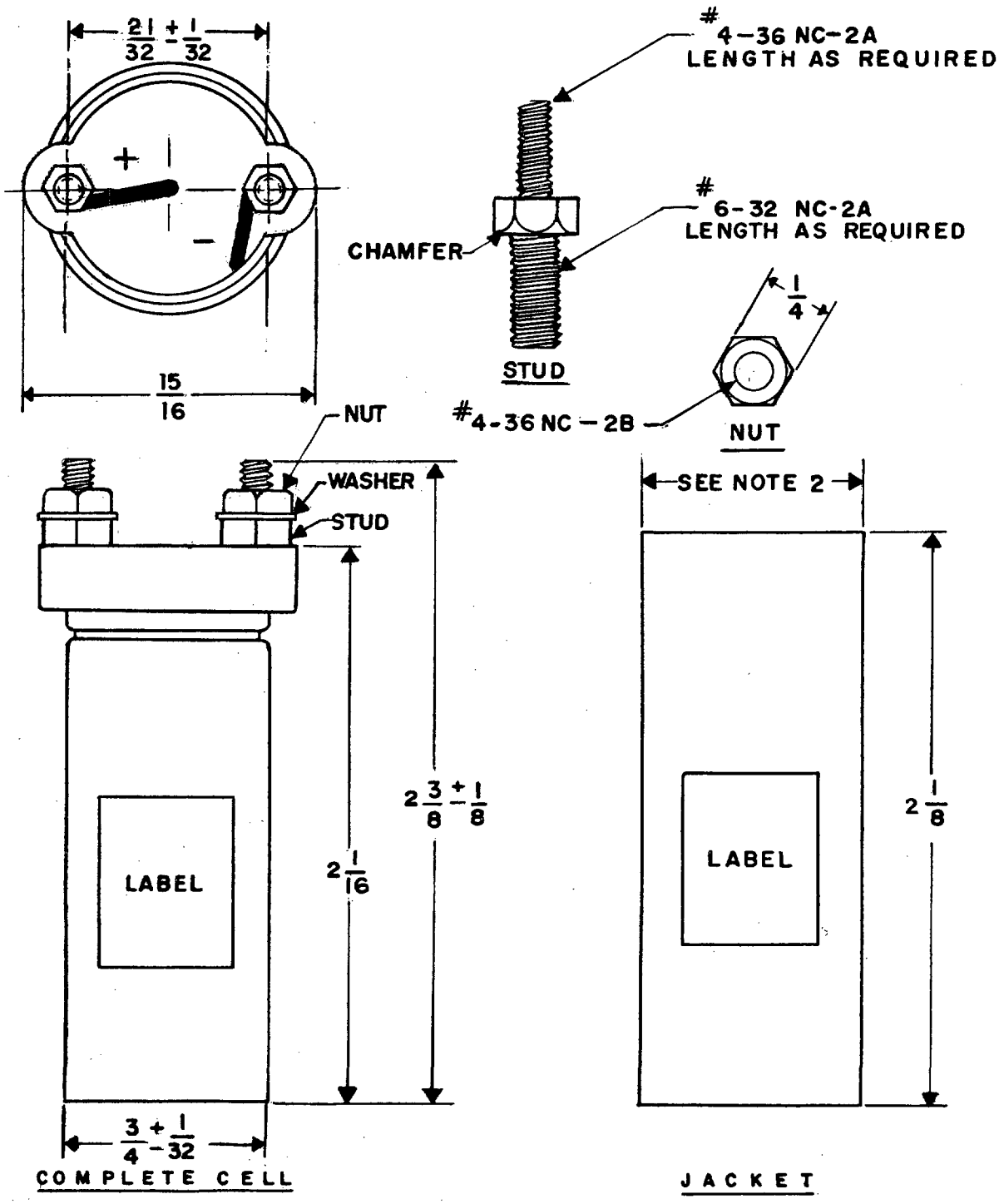
3.3.1 Cell. The cell of the battery generally consists of:

(a) A positive electrode of silver chloride which serves as a positive terminal.

(b) A negative electrode of high-purity zinc which serves as a negative terminal, and as a can housing the constituent parts of the battery.

(c) Electrolyte consisting of one or more metallic chlorides, together with whatever additives that may be necessary to provide the electrical characteristics of the battery types specified herein.

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1. ALL DIMENSIONS IN INCHES, TOLERANCES ARE ±1/16 OR AS SPECIFIED
2. THIS DIMENSION SHALL BE OF A SIZE TO ENABLE THE JACKET TO PUSH-FIT OVER THE COMPLETED CELL. THE JACKET SHALL BE OPEN TOP AND BOTTOM AND EXTEND APPROXIMATELY 1/4 BELOW THE BOTTOM OF THE CELL.

FIGURE 1

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3.3.2 Dimensions. All specified dimensions (see Fig. 1) shall be inclusive of any coating that may be applied, and shall remain within the specified tolerances during storage periods and throughout all tests specified herein.

3.3.3 Threaded parts. All threaded parts shall be in accordance with Handbook H28. Where practical, all threads shall be in conformity with the coarse-thread series. The fine-thread series shall be used only for applications that might show a definite advantage through their use.

3.3.3.1 Engagement of threaded parts. All threaded parts shall engage by at least three full threads.

3.3.4 Weight. The weight of each battery shall be not more than 50 grams.

3.3.5 Voltage.

3.3.5.1 Nominal voltage. The nominal voltage of the battery shall be 0.9 volt.

3.3.5.2 Closed-circuit voltage. The closed-circuit voltage of the battery shall be not less than 0.85 volt nor more than 1.0 volt at the time capacity discharge is initiated (see 4.7.2 and 4.7.5).

3.3.5.3 Test-end voltage. The test-end voltage, during capacity discharge, shall not fall below 0.80 volt for battery types BA-245/U and BA-2245/U. (See 4.7.5)

3.3.6 Terminals. Terminals shall provide a positive means of making and maintaining external electrical connections to the battery. The terminals shall be of the stud and nut type; consisting of a stud with a hexagon-chamferred shoulder, one hexagon nut, and one flat washer. (See Fig. 1) If studs are painted with nuts in place, electrical continuity must be maintained.

3.3.7 Jacket. The jacket shall be a removable cylindrical tube, extending below the bottom of the cell as shown on Figure 1, and shall cover the cylindrical surface of the cell. The jacket may be of metallic or nonmetallic material (see 3.3.7.1 and 3.3.7.2).

3.3.7.1 Metallic jacket. When a metallic jacket is provided, the jacket shall be completely insulated from the cell (see 3.3.1). The exterior surface of the jacket shall have a protective coating to prevent corrosion during storage and throughout all tests specified herein.

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3.3.7.2 Nonmetallic jacket. When a nonmetallic jacket is provided, the jacket shall be held together with a waterproof adhesive in such a manner as to prevent it from coming apart during storage and throughout all tests specified herein.

3.3.7.3 Color of jacket. The color of the exterior surface of the jacket shall be lusterless No. 34087, in accordance with Federal Standard No. 595. The color of the cell shall be the same as the jacket.

3.3.8 Sealing. When the cell is sealed at the terminal end of the jacket with an insulating compound, such a seal shall exclude moisture from the insulating compound without impairing its electrical characteristics, and shall not flow, crack, or become loose at any orientation that may be assumed by the battery under the most severe and test conditions specified herein.

3.4 Waterproofing of jackets. When subjected to waterproofing-of-jackets test, the jacket shall not fall apart and shall show no evidence of water penetration nor separation between layers of jacket material (see 4.7.3).

3.5 Electrolyte leakage. There shall be no evidence of electrolyte leakage on the exterior surface of the cell or the exterior of the battery with the jacket in place at any time prior to or during the storage conditions specified herein. The capacity of any battery that shows evidence of electrolyte leakage (during storage or discharge) before its voltage drops to the test end voltage shall be considered zero. Electrolyte leakage shall be considered to have occurred when moisture appears on the surface or appears on a piece of absorbent paper rubbed on the surface of the battery.

3.6 Vibration. While the battery is being subjected to the vibration test specified in 4.7.4, the voltage under no load shall not fluctuate or fall below 0.85 volts, and there shall be no evidence of electrolyte leakage on any of its external surfaces. After the vibration test is completed, the closed-circuit voltage and capacity discharge shall be as specified in 3.3.5.2 and 3.7, respectively. In addition, there shall be no evidence of loosening, cracking, breaking, tearing, or deformation of the terminals, jacket, cell can, and the seal; and the labeling and marking shall not become illegible.

3.7 Capacity discharge. The batteries shall complete the number hours of service specified in Table II (BA-245/U) or the number hours of service specified in Table IIa (BA-2245/U), before the voltage of the battery falls below the test-end voltage, or before the battery shows evidence of electrolyte leakage (see 3.3.5.3, 3.5 and 4.7.5).

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Table II - (BA-245/U)

Capacity-discharge test	Minimum number of hours service	Method Paragraph
Preliminary temperate discharge (Preproduction)	45	4.7.5.2
Initial temperate discharge	45	4.7.5.3
Delayed temperate discharge	40	4.7.5.4
Temperate discharge after tropical storage	35	4.7.5.5

Table IIa - (BA-2245/U)

Capacity-discharge test	Minimum number of hours service	Method Paragraph
Preliminary temperate discharge (Preproduction)	50	4.7.5.2
Preliminary arctic discharge (Preproduction)	30	4.7.5.2
Initial temperate discharge	50	4.7.5.3
Delayed temperate discharge	40	4.7.5.4
Initial arctic discharge	30	4.7.5.6
Delayed arctic discharge	20	4.7.5.7

3.8 Minimum performance index (MPI). The MPI of a battery shall be computed according to the formulas specified in 3.8.1 and 3.8.2, as applicable.

3.8.1 Battery BA-245/U.

$$MPI = (1 + H)\ (IT + 4DT)$$

Where:

$$H^1 = \frac{TS}{IT} \times 0.3$$

TS = Specified minimum number of hours service on temperate discharge after tropical storage (see Table II).

IT = Specified minimum number of hours service on initial temperate discharge (see Table II).

DT = Specified minimum number of hours service on delayed temperate discharge (see Table II).

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^{1} Only values of H within the range of 0 to 0.3, inclusive, shall be used in computing the MPI. When no minimum requirements are given for temperate discharge after tropical storage, the MPI is calculated on the basis of H = 0, using the formula given above.

EXAMPLE: To calculate the MPI of Battery BA-245/U:

In Table II, the specified minimum number of hours service is:

TS = 35 hours
IT = 45 hours
DT = 40 hours

Substituting these values in the above formulas, the following is obtained:

$$H = \frac{35}{45} \times 0.3 = .231$$

$$MPI = (1 + .231)\left[45 + (4 \times 40)\right] = 252$$

3.8.2 Battery BA-2245/U.

$$MPI = (1 + A)\quad (IT + 4DT)$$

$$A = \left(\frac{IA}{IT}\right) + 4\left(\frac{DA}{DT}\right)$$

IA = Specified minimum number of discharge hours on initial arctic discharge (see Table IIa).

DA = Specified minimum number of discharge hours on delayed arctic discharge (see Table IIa).

EXAMPLE: To calculate the MPI of Battery BA-2245/U: In Table IIa, the specified minimum number of discharge hours is:

IA = 30 hours
DA = 20 hours
IT = 50 hours
DT = 40 hours

Substituting these values in the above formulas, the following is obtained:

$$A = \left(\frac{30}{50}\right) + 4\left(\frac{20}{40}\right) = 2.6$$

$$MPI = (1 + 2.6)\left[50 + (4 \times 40)\right] = 756$$

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3.8.3 Actual performance index (API). The API of a battery shall be computed by substituting in the applicable formula (see 3.8.1 or 3.8.2) the average number of discharge hours obtained from all the sample batteries allocated and subjected to each capacity-discharge test (see Tables II and IIa). The API shall be computed for each shipment lot.

3.8.4 Guaranteed performance index (GPI). The GPI of a battery shall be computed by substituting in the applicable formula (see 3.8.1 or 3.8.2) the number of discharge cycles or hours specified in the contract or order. When GPI is not specified in the contract or order, the MPI value shall be used.

3.9 Labeling and marking. All labeling and marking on the battery shall be clear and legible throughout all of the test specified herein, and shall be in black lettering (see 3.3.7.3). However, when labeling or marking is applied by means of embossing, it may be of the same color as the background. Nonmetallic jackets, except plastic jackets, shall not have the labeling and marking embossed.

3.9.1 Label. The battery jacket and the cell shall have a label imprinted with the following information:

BATTERY, DRY
(Type designation)
(Order number)
(Code)
SUPPLIER'S NAME
(Trade name may also be used)
SUPPLIER'S ADDRESS

EXAMPLE:

BATTERY, DRY
BA-245/U
DAAB05-67-C-9999
0967
JOHN DOE COMPANY
JODOCO
BATTERYVILLE, N.J.

3.9.1.1 Code. The code shall indicate the month and year of fabrication of the battery by means of a 4-digit number in which the last two digits shall indicate the year. Months earlier than the tenth month shall be a single digit preceded by "0".

EXAMPLE: A battery fabricated in March 1960 will bear the code "0360".

A battery fabricated in November 1959 will bear the code "1159".

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When a battery is completed during the last 3 working days of a month or the first 3 working days of a subsequent month, the supplier is permitted to use either month as the date to be coded.

3.9.2 Marking. Terminal polarity and the voltage of 0.9 volt shall be marked on the top, or on the side, or both surfaces of the battery. The polarity markings shall clearly indicate the terminal to which they refer.

3.10 Workmanship. Batteries shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, serviceability, or appearance.

3.10.1 Imperfections. The following imperfections shall be considered as affecting the life and serviceability of the batteries:

(a) Terminals or connections which are loose or corroded to the extent that electrical contact cannot be made and maintained, obstructed with wax or foreign substances which prevent good electrical contact, or which may cause corrosion in storage, or which are fabricated or installed in a manner as to prevent potential open or short circuits.

(b) Potential short circuits due to:

(1) Omitted, damage, or improperly placed insulation.

(2) Electrolyte leakage.

(3) Conducting particles such as zinc filings, soldering-iron scales, loose pieces of solder, and other material.

(c) Cell cans which leak or are distorted; which have flaws such as cracks, blisters, holes, grooves on the interior or exterior surfaces, or sharp dents; or which have large dents below air space.

(d) Seals which are loose, cracked, brittle, low, or raised; which obstruct the use of terminals; which contain holes extending through the entire seal thickness; or which show evidence of leakage or plastic flow.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the

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supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government. Inspection equipment furnished by the Government, for performing certain inspection, will be calibrated and maintained by the Government.

4.1.1.1 Instrument accuracy.

4.1.1.1.1 Voltmeters and ammeters. Voltmeters and ammeters used in testing batteries shall be accurate within ± 1 percent of the full-scale value. The voltmeter and ammeter ranges used shall be such that all readings are taken on the upper half of the scale. The sensitivity of voltmeters shall be at least 1,000 ohms per volt.

4.1.1.1.2 Resistor tolerances. In all tests involving discharge through a resistance, such resistance shall be accurate within 0.5 percent. In determining the resistance used as a test load, the resistance of all continuously operating voltmeters shall be considered as part of the specified total resistance.

4.1.1.1.3 Timing. Timing equipment shall be accurate within 0.5 percent.

4.2 Classification of inspection. The examination and testing of batteries shall be classified as follows:

(a) Materials and components inspection (see 4.3).

(b) Preproduction inspection (see 4.5).

(c) Quality conformance inspection (see 4.6).

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.6.1).

(2) Quality conformance inspection of preparation for delivery (see 4.7.6).

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4.3 Materials and components inspection. Material and components inspection shall consist of verification by certification that the component materials listed in Table I, used in fabricating the batteries, are in accordance with the applicable specifications and requirements prior to such fabrication. When certification, either from the source or as to the test results of a particular lot of component materials, has been furnished to the Government on a contract or order, no further certification for that lot of component materials will be required for subsequent contracts or order.

4.4 Inspection conditions. Unless otherwise specified herein, all examinations and tests shall be performed at a temperature of 80° ± 20°F., ambient relative humidity, and atmospheric pressure.

4.5 Preproduction inspection. As soon as possible after award of contract and prior to production, preproduction inspection as specified in 4.5.1 to 4.5.1.3, inclusive, will be performed by the supplier. Preproduction inspection as specified in 4.5.2 will be performed by the Government.

4.5.1 By the supplier. Prior to preproduction inspection, the supplier shall perform component-materials inspection (see 4.3). Preproduction inspection shall consist of fabrication of sample batteries (see 4.5.1.1) and shipment of sample batteries to a Government laboratory (see 4.5.1.2).

4.5.1.1 Fabrication of sample batteries. Unless otherwise specified (see 6.2) the supplier shall fabricate 30 sample batteries (BA-245/U) or 45 sample batteries (BA-2245/U), as applicable, of each type on a contract or order, each constituting a sample lot.

4.5.1.2 Marking and shipping of sample batteries. A label as specified in 3.9 marked "PREPRODUCTION SAMPLE" shall be securely attached to each sample battery. The batteries shall be packaged as specified in 5.1 and shipped to:

Commanding General
U. S. Army Electronics Command
Procurement and Production Directorate
Squier Area, Building T-292
ATTN: AMSEL-PP-ET-3
Fort Monmouth, New Jersey 07703

One additional sample battery shall be furnished the Government, at the place of manufacture, to be used as a standard for comparative purposes when necessary.

4.5.1.3 Cost. Sample batteries supplied for preproduction inspection, and if required for reinspection, shall be supplied in addition to the number called for in the contract or order and shall be furnished at no additional cost to the Government.

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4.5.2 By the Government. Upon receipt at the Government laboratory of batteries (BA-245/U) marked "PREPRODUCTION SAMPLE", the 30 sample batteries will be divided into three groups: Group I of five batteries, Group II, fifteen batteries and Group III of ten batteries. Upon receipt at the Government laboratory of batteries (BA-2245/U) marked "PREPRODUCTION SAMPLE", the 45 sample batteries will be divided into three groups: Group I of five batteries, Group II of twenty batteries and Group III of twenty batteries. These groups of batteries will be subjected to the following examination and tests, in the order listed:

Group I (BA-245/U - BA-2245/U)

Vibration (see 4.7.4)
Closed circuit voltage (see 4.7.2)
Waterproofing of jackets (see 4.7.3)
Visual and mechanical examination (see 4.7.1)

Group II (BA-245/U)

Vibration (see 4.7.4)
Visual and mechanical examination (see 4.7.1)
Preliminary temperate discharge (see 4.7.5.2)

Group II (BA-2245/U)

Vibration (see 4.7.4)
Visual and mechanical examination (see 4.7.1)
Preliminary temperate discharge (see 4.7.5.2)
Preliminary arctic discharge (see 4.7.5.2)

Group III (BA-245/U - BA-2245/U)

Electrolyte leakage test (see 4.7.5.8)

4.5.2.1 Failure. If any sample battery fails the Government preproduction inspection (see 4.5.2), the supplier shall immediately make the necessary remedial changes, and may be required to submit additional preproduction sample batteries for reinspection. These additional sample batteries shall be accompanied by a description of the remedial changes made to correct the faults in the previous submission. Approval will be given only when an entire set of sample batteries submitted have been found satisfactory.

4.5.3 Start of production. The supplier shall not engage in production until the certification furnished for the component materials used in the fabrication of batteries, and the sample batteries submitted for preproduction inspection have been approved. Any production prior to approval shall be solely at the supplier's risk.

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4.6 Quality conformance inspection.

4.6.1 Quality conformance inspection of equipment before preparation for delivery. This inspection shall consist of groups A and B inspection, to be performed by the contractor, and group C inspection to be performed by the Government at their test facilities.

4.6.2 Definitions.

4.6.2.1 Contract lot. The contract lot is the total number of all batteries of any one type, delivered in one or more shipment lots, on any one contract or order.

4.6.2.2 Shipment lot. The shipment lot is the quantity of batteries (exclusive of the number of batteries required as samples) produced in one month, of any one type, at any one place of manufacture, on any one contract or order.

4.6.2.3 Production lot. The production lot shall be as defined in Standard MIL-STD-105.

4.6.3 Resubmitted lot. If any inspected lot is rejected, the supplier may withdraw the lot, rework it to correct the defects, or screen out the defective units, and again inspect it as applicable. Such lots shall be kept separate from new lots and shall be clearly identified as reinspected lots. Such lots shall be reinspected, using tightened inspection.

4.6.4 Group A in-plant inspection. This inspection including sampling, shall conform to Table III and the inspection procedures of Standard MIL-STD-105, using general inspection level II.

Table III - Group A in-plant inspection

Examinations and test	Rqt Para.	Method of Test	AQL[1]	
			Major	Minor
Visual and mechanical examination (External)	3.10[2]	4.7.1	1.0%	4.0%
Closed circuit voltage test	3.3.5.2	4.7.2	1.0%	

[1] Acceptable quality level (percent defective).

[2] Imperfections listed under paragraph 3.10.1 are considered Major defects. Any defect not listed under paragraph 3.10.1 which will affect life or serviceability of the battery shall be considered a Major defect. Minor defects shall be considered as those defects that will not affect life and serviceability of the batteries.

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4.6.5 Group B in-plant inspection. Group B inspection shall consist of the electrolyte leakage test (4.7.5.8). This inspection including sampling shall conform to the procedures for small sampling of Standard MIL-STD-105, using the special inspection levels. The AQL shall be 4.0% defective and the inspection level shall be S-4 for normal, tightened and reduced inspection.

4.6.6 Group C inspection. Group C inspection will be performed by the Government at a Government laboratory (see 4.5.1.2) and will consist of tests specified in Table V (see 4.6.6.5). A different set of sample batteries shall be selected for each test. Shipment of batteries on the contract or order shall not be held up pending the completion of Group C inspection (see 6.4).

4.6.6.1 Selection of group C sample batteries. Sample batteries shall be selected at random from each shipment lot for group C inspection as specified in Table IV.

Table IV - Sample batteries for group C inspection

Contract-lot size	Sample size for each capacity test from contract lot
Under 500[1]	12
500 to 2,500	36
2,501 to 5,000	40
5,001 to 10,000	45
10,001 to 25,000	50
Over 25,000	60

[1] See 4.6.6.5

4.6.6.2 Sample size for each shipment lot. The number of sample batteries selected from each shipment lot shall be in the same ratio to the total number specified in Table IV as the shipment lot is to the contract lot.

4.6.6.3 Marking and packaging of sample batteries. The selected sample batteries shall be unit packaged as specified in 5.1.1 or 5.1.2, as applicable. Any void spaces in the package shall be filled with filler or padding material or dummy batteries. Each package shall be marked "RESERVED FOR GROUP C INSPECTION. DO NOT OPEN UNTIL COMPLETION OF STORAGE PERIOD".

4.6.6.4 Shipment of sample batteries. Sample batteries packaged and marked as specified in 4.6.6.3 shall be shipped to the Government laboratory specified in 4.5.1.2, for group C inspection at no additional cost to the Government.

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Table V - Group C inspection

Test	Applicable to battery type	Reqt Para.	Method Para.
Capacity discharge:		3.7	4.7.5
Initial temperate discharge	BA-245/U and BA-2245/U	---	4.7.5.3
Delayed temperate discharge	BA-245/U and BA-2245/U	---	4.7.5.4
Temperate discharge after tropical storage	BA-245/U	---	4.7.5.5
Initial arctic discharge	BA-2245/U	---	4.7.5.6
Delayed arctic discharge	BA-2245/U	---	4.7.5.7

4.6.6.5 Contract lots smaller than 500 batteries. When contract-lot sizes smaller than 500 batteries are involved, noncompliance specified in 4.6.6.6 shall not apply. However, the 12 sample batteries selected (see Table IV) shall be subjected to visual and mechanical examination and preliminary-temperature-discharge test only (see 4.7.1 and 4.7.5.2).

4.6.6.6 Noncompliance. If the API is less than the GPI, calculated as specified in 3.8.3 and 3.8.4, respectively, the shipment lot represented by the sample batteries shall be considered as not complying with the requirements of this specification and an adjustment shall be made as specified in the contract or order (see 6.5).

4.7 Methods of examination and tests.

4.7.1 Visual and mechanical examination. Batteries shall be visually examined as specified in Table VI to verify that they are in accordance with the applicable requirements. Internal examinations shall be made at each applicable point in the process, and as specified herein. In addition to the external examinations listed in Table VI, at the conclusion of each test condition, the sample batteries shall be examined for conformance to the following requirements:

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Metallic jackets (corrosion of) (see Table I and 3.2.1)
Nonmetallic jackets (condition of) (see 3.3.7.2)
Sealing (see 3.3.8)
Electrolyte leakage (see 3.5)
Labeling and marking (see 3.9 to 3.9.2 incl.)
Workmanship (see 3.10)

Table VI - Visual and mechanical examination (external)

Requirement	Requirement Para.
Design and construction	3.3
Dimensions	3.3.2
Weight	3.3.4
Terminals	3.3.6
Color of jacket	3.3.7.3
Labeling and marking	3.9 to 3.9.2, incl.

4.7.2 Closed-circuit voltage. The battery shall be stored for 24 hours at a temperature of 70° ± 2 F., and relative humidity of 50 ± 15 percent. While the battery is still at these conditions the closed-circuit voltage shall be measured when the battery is loaded with a fixed resistor of 115 ohms (see 3.3.5.2).

4.7.3 Waterproofing of jackets. The battery with the jacket in place, shall be immersed in water to within ¼ inch of the top of the jacket for 48 hours. The temperature of the water shall be maintained at 110°F ± 2°F. during immersion. The battery shall then be examined for the condition of the jacket (see 3.4).

4.7.4 Vibration. The battery shall be rigidly clamped to the platform of a vibration machine in a manner approximating as closely as practicable the manner in which the battery is clamped when in use. A simple harmonic motion having an amplitude of 0.03 inch (0.06 total maximum excursion) shall be applied. The frequency of vibration shall be varied at the rate of 1 cycle per second per minute between the limits of 10 and 55 cycles per second. The entire range of frequencies, ascending and descending, shall be traversed in not less than 90 minutes nor more than 100 minutes. Each battery shall be vibrated in two mutually perpendicular directions, one of which shall be perpendicular to the terminal face. Throughout the vibration periods, observations shall be made of voltage under no load and evidence of electrolyte leakage. After completion of the test, the battery shall be visually examined (see 3.6 and 4.7.1).

4.7.5 Capacity discharge. Prior to discharge, the sample batteries in unit package, as received, shall be placed in storage at the temperature and relative humidity conditions for the period specified in 4.7.5.1 to 4.7.5.7, inclusive as

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applicable. Upon completion of the storage period, each battery shall be subjected to visual and mechanical examination (see 4.7.1) and then short circuited until a current value of at least 150 milliamperes (ma) is built up. Then each battery shall be discharged through a fixed resistance of 115 ohms ± .05% continuously. The discharge shall continue at the temperature and relative humidity conditions specified in 4.7.5.1 to 4.7.5.7, inclusive, as applicable, until the voltage falls below the test-end voltage specified in paragraph 3.3.5.3 or there is evidence of electrolyte leakage.

4.7.5.1 Ambient storage and discharge conditions. The ambient conditions specified in Table VII, shall prevail during entire storage and discharge periods. Normal conditions shall be maintained insofar as possible. Deviations from normal conditions are permitted, provided that the extreme conditions specified in Table VII do not exist for more than 5 percent (cumulative) of the specified storage or discharge periods.

Table VII - Ambient storage and discharge conditions

Kind of Storage or discharge	Normal conditions		Extreme conditions	
	Temperature °F	Relative humidity (Percent)	Temperature °F	Relative humidity (Percent)
Temperature storage	70 ± 5	50 ± 15	60 to 65, incl, and 75 to 80, incl.	25 to 35, Incl, and 65 to 70, incl.
Tropical storage	113^{-8}_{+2}	50 ± 15	88 to 105, incl, and 115 to 118, incl.	25 to 35, incl, and 65 to 75, incl.
Arctic storage	-20 ± 2	----	(1)	-----
Temperature discharge	70 ± 2	50 ± 15	65 to 68, incl, and 72 to 80, incl.	25 to 35, incl, and 65 to 75, incl.
Arctic discharge	-20 ± 2	----	(1)	-----

(1) Deviation from normal conditions are not permitted.

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4.7.5.2 Preliminary temperate and arctic discharge. The sample batteries shall be placed in either temperate or arctic storage (as applicable) for 48 hours and then given a temperate and/or arctic discharge.

4.7.5.3 Initial temperate discharge. The sample batteries, as received, shall be placed in temperate storage for 1 month and then given a temperate discharge (see 4.7.5 and 4.7.5.1).

4.7.5.4 Delayed temperate discharge. The sample batteries, as received, shall be placed in temperate storage for 12 months and then given a temperate discharge (see 4.7.5 and 4.7.5.1).

4.7.5.5 Temperate discharge after tropical storage. The sample batteries, as received, shall be placed in tropical storage for 3 months. Following the tropical storage period, the sample batteries shall be further stored at temperate storage for 48 hours and then given a temperate discharge. (See 4.7.5 and 4.7.5.1)

4.7.5.6 Initial arctic discharge. The sample batteries, as received, shall be placed in temperate storage for 3 months. Following the temperate-storage period, the sample batteries shall be further stored at arctic storage for 48 hours and then given an arctic discharge. (See 4.7.5 and 4.7.5.1)

4.7.5.7 Delayed arctic discharge. The sample batteries, as received, shall be placed in temperate storage for 12 months. Following the temperate-storage period, the sample batteries shall be further stored at arctic storage for 48 hours and then given an arctic discharge. (See 4.7.5 and 4.7.5.1)

4.7.5.8 Electrolyte leakage test. Sample batteries shall be discharged through a 115 ohm (±0.5%) load, continuously for fifty (50) hours and then stored for 15 days. The storage and discharge shall be performed at inspection conditions specified in paragraph 4.4. During the 15 day storage period, each battery shall be examined daily for evidence of electrolyte leakage on the external surfaces of the battery. (See 3.5)

4.7.6 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Packaging shall be level A or C as specified in the contract or order (see 6.2(e)).

5.1.1 Level A.

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5.1.1.1 Cleaning. Each battery shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each battery shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. Batteries shall be packaged individually in accordance with method IC-1 of MIL-P-116. The heat-sealed bag shall be fabricated from material conforming to L-P-378.

5.1.1.5 Intermediate packaging. Twenty-five unit packages shall be placed in a close-fitting box conforming to PPP-B-566, variety 2, or PPP-B-676, variety 2. Closure shall be as specified in the appendix to the box specification. The interior of the box shall be fitted with slotted partitions to provide snug protection to each battery.

5.1.2 Level C. Batteries shall be packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified. (See 6.2(e))

5.2.1 Level A.

5.2.1.1 A quantity of batteries, of the same type number, packaged as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-636, type CF, class weather-resistant special requirements use. Box closure shall be in accordance with the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. A quantity of batteries of the same type number, packed as specified in 5.2.1.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s. The load shall be bonded to the pallet by strapping.

5.2.1.3 Less than palletized load. When quantities per destination are less than a pallet load, containers specified in 5.2.1.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, type IV as specified in the appendix to the box specification. No further packing shall be required.

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5.2.2 Level B. Unless otherwise specified, a quantity of batteries of the same type number, packaged as in 5.1, shall be packed as specified in 5.2.1.

5.2.3 Level C. Batteries of the same type number, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that pallets, fiberboard boxes, and fiberboard caps shall be of the type, size, and kind commonly used for the purpose and shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.2.4 General. Shipping containers for a battery type shall be uniform in shape, size, style, and kind. Shipping containers of minimum cube and weight consistent with the protection prescribed are required, and shall contain identical quantities of identical items.

5.3 Marking. Interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Battery type BA-2245/U only: Each battery shall have a printed wrap-around label, on Olive Drab paper, in black Bold Face letters, outside of the polyethylene bag and held in place with an elastic band, with the following information: This battery is designed for use at low temperature (below 20°F) but is also satisfactory for use at normal temperatures. Store at lowest temperature practicable; temperature in excess of 70°F will cause rapid deterioration. For maximum service at low temperature, whenever feasible, warm battery to approximately 70°F before using. When used cold, battery will be only partially discharged but if warmed, will give additional service.

5.3.2 In addition to any special markings required by the contract or order each shipping container shall be marked:

PERISHABLE
STORE UNDER REFRIGERATION
TO PREVENT DETERIORATION

6. NOTES

6.1 Intended use. The batteries covered by this specification are intended for use as the power source for blasting-circuit galvanometers.

6.2 Ordering data. Procurement documents should specify the following:

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(a) Title, number, and date of this specification.

(b) Complete type designation. (See 1.2.1)

(c) Number of sample batteries if other than that specified in 4.5.1.1.

(d) Date for notice of availability for shipment.

(e) Levels of preservation and packaging and packing and applicable marking. (See Section 5)

(f) Number of unit packages if other than that specified in 5.1.1.

6.3 Awarding of contracts. Contracts will be awarded only to bidders who guarantee to meet the MPI (see 3.8) or better, and the requirements of this specification. Failure on a prior contract, of a supplier's particular battery type to meet any of the performance requirements of this specification, will be adequate cause for rejection of bids on that particular battery type until the supplier submits certified data proving that:

(a) Action has been taken to eliminate the cause of failure.

(b) The battery meets all performance requirements of this specification.

6.4 Group C inspection. Group C inspection is performed by the Government to obtain information regarding the performance of batteries selected from production lots, when tested under standard conditions. The results of these tests are not used by the Government to reach a decision on acceptance or rejection of any lot batteries offered as a delivery against a contract or order, but are used as a basis for adjustment. (See 4.6.6.2)

6.5 Adjustment in the event of failure to meet GPI. In the event API, calculated as specified in 3.8.3, is less than 80 percent of GPI (see 3.8.4) or in the event weighted average (WA) (see 6.5.3) is 80 percent or more, but less than 95 percent of GPI, and provided the supplier is notified in writing of the results within 6 months after completion of group C inspection, adjustments should be made as specified in 6.5.1.

6.5.1 Adjustments. Adjustments should be made in cash, or if mutually agreeable between the supplier and the Government, in additional batteries. All such additional batteries should conform to all requirements of this specification, and should be subject to adjustment as specified herein; however, these adjustments should be made in cash only.

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6.5.2 Extent of adjustment on shipment lot. On those shipment lots in which the API is less than 80 percent of the GPI, the adjustment should be determined by the following formula for each such lot:

$$\text{Value of adjustment} = M \frac{(GPI - API)^2}{API}$$

Where: M = The dollar value of the batteries comprising the shipment lot. In no case should any shipment - lot adjustment (whether in cash, additional batteries, or a combination of cash and additional batteries) exceed 70 percent of the contract dollar value of the specific shipment lot represented. (See 4.6.2.2)

6.5.3 Extent of adjustment on WA. On those shipment lots in which the API is 80 percent or more, but less than 95 percent of the GPI, the adjustment should be determined by the following formula for all such lots:

$$\text{Value of adjustment} = N \frac{(GPI - WA)^2}{WA}$$

Where: N = The total-contract dollar value of the batteries minus the total dollar value of the batteries in all the shipment lots which were subject to shipment-lot adjustment (see 6.5.2). In the event there are no shipment-lot adjustments, the N value is the total-contract dollar value of the batteries.

WA = The weighted average is computed as follows:

(a) On each shipment lot in which the API is 80 percent or more of the GPI, the number of batteries in each such shipment lot is multiplied by its API.

(b) These products are added.

(c) The resulting sum is divided by the total number of batteries on these shipment lots.

No adjustment should result if the computed WA is 95 percent or more of the GPI.

6.5.4 Total adjustment.

6.5.4.1 In cash. The total adjustment made in cash for batteries on contract should be equal to the sum of all shipment-lot adjustments in accordance with 6.6.2, plus the WA adjustment in accordance with 6.5.3.

6.5.4.2 In batteries. The total adjustment made in batteries should be the quantity of batteries to be furnished as determined by dividing the total adjustment in cash (see 6.5.4.1) by the contract price of each battery.

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NOTE: Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians
Army - EL
Navy - SH
Air Force - 82

Review activities
Army - SM
Navy - SH, AS
Air Force - 82, 85

User activities
Army - AV, MI
Navy - MC

Preparing activity
ARMY - EL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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ESC-FM 2011-66

To detach this form, cut along this line

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